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L2

Solution: **D**

Since the electrons and protons are moving in opposite directions, the total current is the sum of the current due to both charges

$$\begin{aligned} I &= \frac{N_e q_e}{t} + \frac{N_p q_p}{t} \\ &= \left(\frac{7 \times 10^{18}}{1} + \frac{2 \times 10^{18}}{1} \right) \times (1.6 \times 10^{-19}) \\ &= 1.44 \text{ A} \end{aligned}$$

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Solution: **B**